Afaq Virk

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Education

Carleton University *Sep 2024–Apr 2027*
B.Comp.Sc. (Honours), Industrial Applications Stream Ottawa, ON
Minor in Economics | **CGPA: 11.9/12.0**

Work Experience

Shopify *Aug 2024–Present*
Software Engineering Intern (Analytics) Ottawa, ON

- Built GCP-backed retrieval infrastructure and developer tooling that ingested embeddings from a six-figure corpus, enabling analytics agents to automate research, surface data insights, and execute recurring workflows.
- Scaled LLM inference and evaluation across Airflow DAGs and Apache Flink pipelines; added regression tracking, rubric-based judges, human annotation, typed schemas, and data contracts to improve quality and reliability across multiple AI features.
- Expanded ShopifyQL in Rust to unlock richer metric exploration and maintained merchant-facing React dashboards to improve analytics and visualization reliability.
- Selected 1 of 30 nationally for Shopify's Dev Degree program with full tuition sponsorship, completed an 8-month training path in Rails, React, and TypeScript, and shipped a real-time React Native storefront prototype in under 24 hours.

MathemaTech *May 2021–Present*
Founder, Lead Educator Remote

- Secured a $3,000 Ontario Ministry grant to launch the business, grew it to $3,000 MRR, and delivered 3,800+ hours of one-on-one mathematics, computer science, and physics tutoring to 150+ secondary and university students in Canada and internationally.
- Created and marketed educational content reaching 700,000+ students and educators and generating 2M+ monthly social impressions; building AI tools to automate practice generation and student tracking.

Upper Canada District School Board *Aug 2023–Jul 2024*
Member, Board of Trustees (Student Trustee) Brockville, ON

- Elected to represent 27,000+ students; led board discussions on STEM, classroom AI, and pedagogy modernization and oversaw $6,000+ in fundraising across 21 schools for CHEO.
- Organized three annual conferences with approximately $75,000 average budgets for about 80 trustees from 35+ school boards and coordinated senior stakeholders, including Ontario's Minister of Education.

Selected Projects

Donair Feb 2026
TypeScript, Next.js, Supabase (Postgres + pgvector), Gemini, Solana Pay

- Built an agentic crowdfunding platform with semantic donor discovery across 110,000+ historical records, weighted capacity, affinity, and timeliness scoring, and real-time Solana Pay flows; secured $8,000 in funding from Carleton University through the Nicol Ventures platform and won the Talsom Sponsor Challenge and Front Row Ventures Investor Pitch Award.

LoRaWATsGoingOn Jan 2026
Python, FastAPI, SQLite, LoRaWAN (ChirpStack)

- Built a ChirpStack-powered device monitor with device-health analytics, anomaly annotations, and gateway navigation; won 1st place at the SSi Canada x Carleton Computer Networks Hackathon.

K.A.R.E.N Jan 2026
SystemVerilog (FPGA), Tcl, Python, Vivado; Gemini; ElevenLabs

- Built an FPGA assistant with HDMI video and keyboard I/O, a constrained LLM-to-hardware command vocabulary for safe deterministic execution, and synchronized voice input, voice cues, and waveform display.

elECHOlocation Oct 2023
Arduino, C++, ultrasonic sensors

- Engineered and experimentally evaluated an ultrasonic navigation aid for visually impaired users; earned $6,000+ in awards, including Gold, Outstanding Research, and the CAP Physics Prize, and represented Canada at MILSET Mexico.

Skills & Technologies

Languages: Rust, Python, C, C++, C#, Java, TypeScript, SQL, SystemVerilog
Frameworks: React, Next.js, FastAPI, Rails, Apache Flink, LangChain, DSPy
Data/Infra: PostgreSQL (pgvector), BigQuery, Airflow, REST APIs, WebSockets
ML/AI: LLM evaluation, RAG, agent design, inference orchestration
Systems/Hardware: LoRaWAN, embedded systems, FPGA digital logic, HDMI/video timing
Tools: Git, Docker, Linux, GCP, Vivado/Tcl, Blender